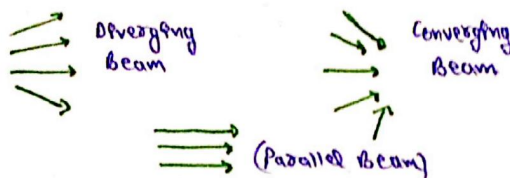


Ray → Very thin beam of light.
→ It is a theoretical concept.

Beam → Bundle of several light rays.



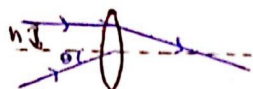
Types of Light Rays :-



(Parallel Rays)

→ Light rays very close to principal axis and almost parallel.

($h, \theta \rightarrow$ negligible)



(Marginal Rays)

→ Light rays far from principal axis and not parallel to P.A.

($h, \theta \rightarrow$ significant)

Reflection of Light :-

→ Bouncing back of light in same medium.



Angle of incidence = Angle of reflection
 $\angle i = \angle r$

2) Incident ray, reflected ray and normal, all lie in the same plane.

$$\vec{e}_2 = \vec{e}_1 - 2(\vec{e}_1 \cdot \vec{n})\vec{n}$$

Also, $[\vec{e}_1 \vec{e}_2 \vec{n}] = 0$

Types of Reflection

Regular Reflection

Diffused Reflection

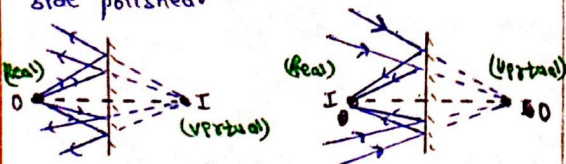


(Light observed in all dirⁿ)

Image Formation by plane mirror :-

Object P.O.I of incident rays
Image P.O.I of reflected rays

→ Plane mirror is a flat glass having one side polished.



1) Object distance = Image distance from mirror

2) Plane mirror does not change nature of beam.

20

Ray Optics - 1

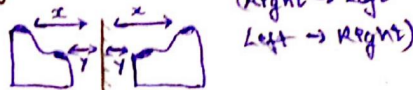
Converging incident beam → converging reflected beam

Diverging incident beam → Diverging reflected beam

1) Nature of object is opposite to that of Image.

object → Image
Real → Virtual
Virtual → Real

1) Plane mirror form Lateral Inversion Image.



Field of View :-

→ To get field of view of an observer / Image, extend lines to extreme ends of mirror.

→ If observer & Image field of view overlap, then Image will be visible.

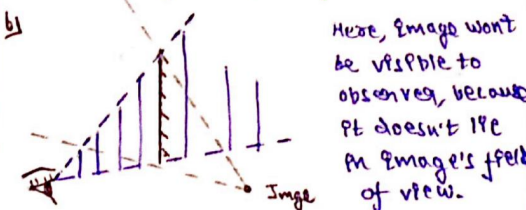
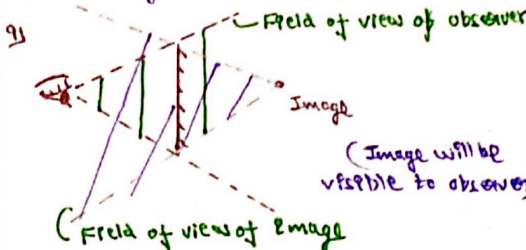
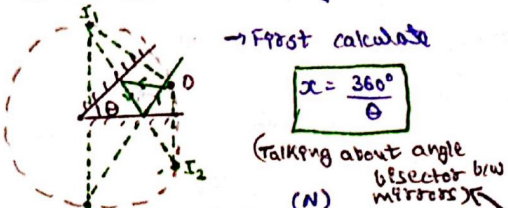


Image formation by inclined plane mirrors :-

→ All Images would lie on a circle whose centre would be P.O.I of both mirrors.



→ First calculate

$$\alpha = \frac{360^\circ}{\theta}$$

(Talking about angle bisector b/w mirrors)

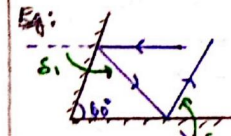
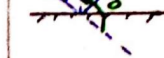
No. of Images = $\alpha - 1$

even → object at bisector, $N = \alpha - 1$
odd → object not at bisector, $N = \alpha$

Fractional Write α as $(2n + f)$ where $f \in (0, 1)$, then $N = 2n$

Deviation due to plane mirror :-

$$\delta = \pi - 2\theta = \pi - 2\delta$$



$$\delta_{net} = \delta_1 + \delta_2$$

By convention,

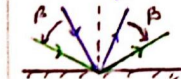
Anticlockwise angle $\Rightarrow +ve$

Clockwise angle $\Rightarrow -ve$

Rotation of Plane Mirror :-

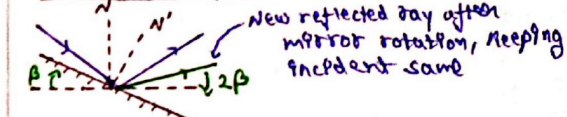
1) Mirror is fixed $\Rightarrow$

If incident ray rotated by β in ACW sense, the reflected ray rotates by β in CW sense



$$\omega_{incident} = -\omega_{reflected}$$

2) Mirror is not fixed & rotates $\Rightarrow$

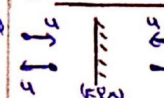


If mirror rotated β in CW sense, then reflected ray rotates 2β in CW sense

$$2\omega_{mirror} = \omega_{reflected}$$

Velocity of Image formed by Plane Mirror :-

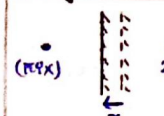
1) Mirror is Fixed $\Rightarrow$



$$\vec{v}_{image} = -\vec{v}_{object}$$

$$|\vec{v}_{image}| = |\vec{v}_{object}|$$

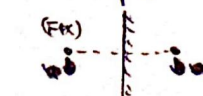
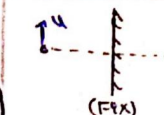
2) Object is Fixed $\Rightarrow$



$$\vec{v}_{image} = 2\vec{v}_{mirror}$$

(Image & mirror move in same dirⁿ)

3) Parallel motion $\Rightarrow$

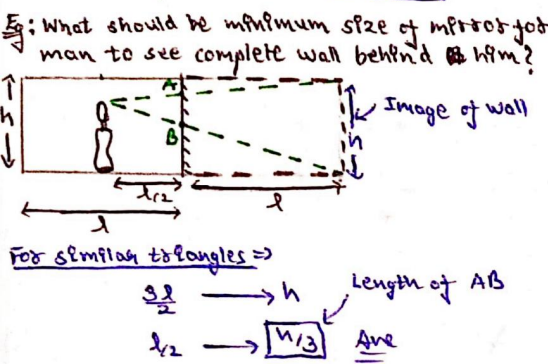
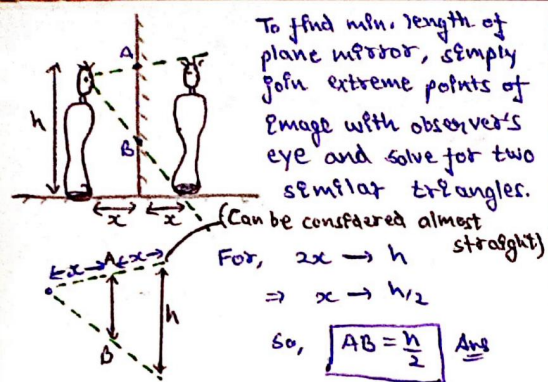


Due to motion of mirror, no movement in Image is observed.

Minimum size of plane mirror :-

→ For a man to see his complete image in a plane mirror, the length of mirror should be half of man's height.

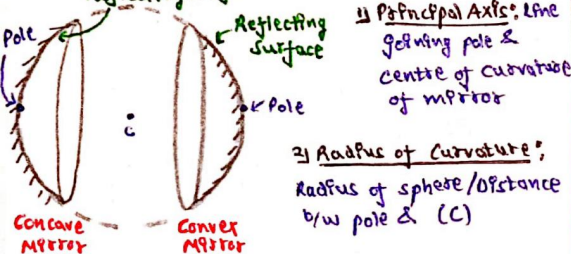
$$l_{mirror} = \frac{l_{man}}{2}$$



#Spherical Mirrors:

→ Curved mirrors are a part of an actual sphere of Radius of curvature R.

Reflecting surf.



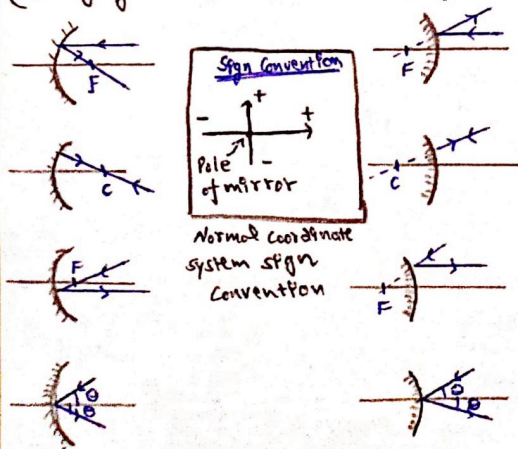
3) Aperture: Distance b/w farthest points of a spherical mirror.



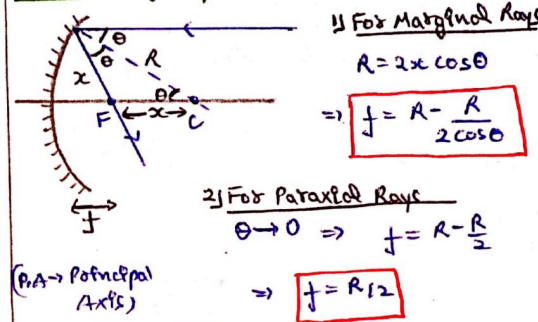
Reflection by Spherical Mirrors $\Rightarrow$

Concave Mirror (Converging Mirror)

Convex Mirror (Diverging Mirror)



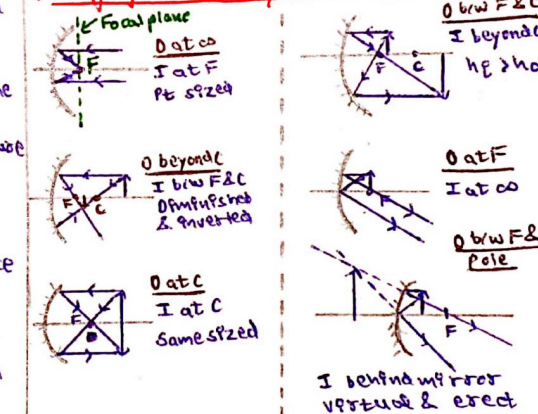
Focal point for spherical mirrors $\Rightarrow$



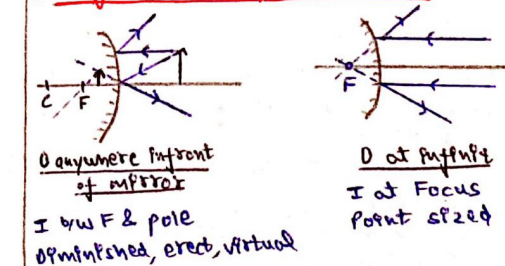
NOTE: Throughout Geometrical Optics

- 1) Real object KP agr Real Image hat, toh woh P.A. ke opp. side hogi (Inverted)
- 2) Real object KP agr virtual Image hat, toh woh P.A. ke same side hogi
- 3) Object aur Image $\Rightarrow$ Same side of P.A. ka nature same hai
- Object aur Image $\Rightarrow$ Opp side of P.A. ka nature opp hai

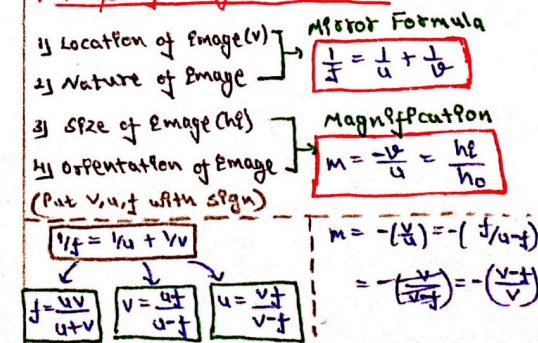
#Image formation by Concave Mirror:



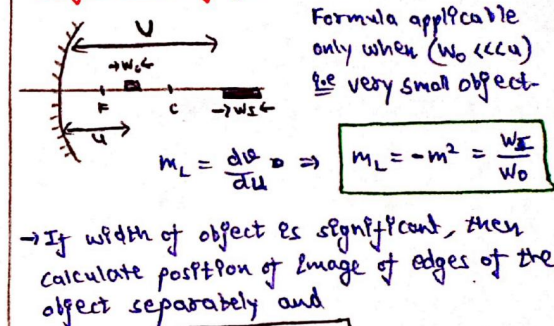
#Image Formation by Convex Mirror:



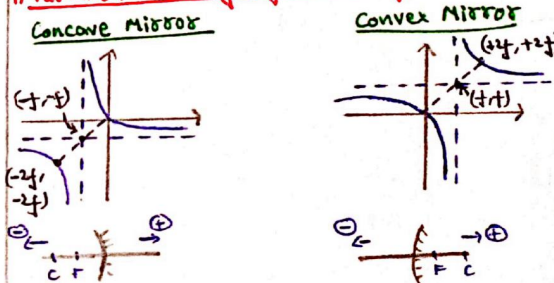
#Analysis of Image Parameters:



#Longitudinal Magnification:

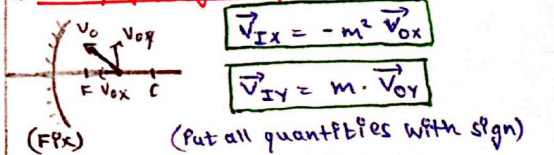


#Variation Curve of Object & Image Distances:



If orientation of mirrors is exchanged, then graphs will also exchange.

#Velocity of Image in Spherical Mirror:



In case if mirror is also moving, then use object velocities ($\vec{v}_o$ & $\vec{v}_m$) w.r.t mirror.

#Power of a Mirror:

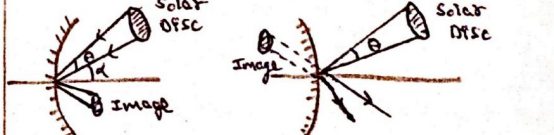
→ Measure of the ability of a mirror to converge or diverge the light rays.

$P = -\frac{1}{f}$ Unit: Diopter (D)

Always put 'f' in metre.

#Image formation of Distant object:

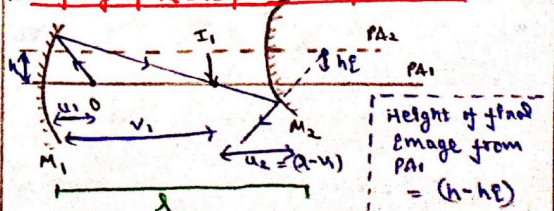
→ Treat every planet/sun as a disc only, because only that portion is visible.



The diameter of image in both cases is given by

$d_i = f \theta$

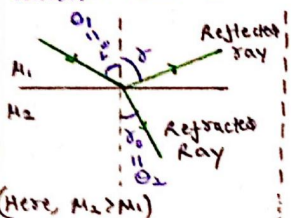
#Shifting of Principal Axis of Mirror:



Refraction of Light:

When light travels from one medium to another, some of it is reflected back into the same med. while rest is refracted into another medium.

$$[M = \frac{c}{v}]$$



Laws of Refraction:-

1) Incident Ray, Normal & Refracted Ray all lie in the same plane.

2) Snell's law $\Rightarrow$

$$n_1 \sin \theta_1 = \text{const.}$$

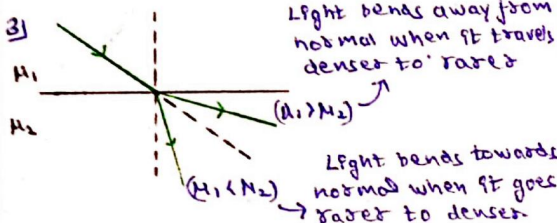
$$\Rightarrow n_1 \sin \theta_1 = n_2 \sin \theta_2$$

So, from here,

$$n_2 = n_{21} = \frac{n_2}{n_1} = \frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

[Relative refractive index of med 2 w.r.t 1]

3)



Light bends away from normal when it travels denser to rarer

Light bends towards normal when it goes rarer to denser

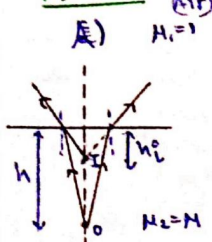
Image formation due to Refraction on plane surface:

Refraction Formula $\Rightarrow$

$$\frac{n_1}{u} = \frac{n_2}{v}$$

$n_1 \rightarrow$ RI of incident ray
 $n_2 \rightarrow$ RI of refracted ray
 $u \rightarrow$ Position of object from interface
 $v \rightarrow$ Position of image from interface.

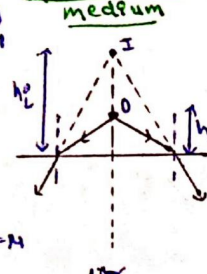
Object in Denser medium



$$h_e = \frac{h}{M}$$

$$\text{Shift} = h \left(1 - \frac{1}{M}\right)$$

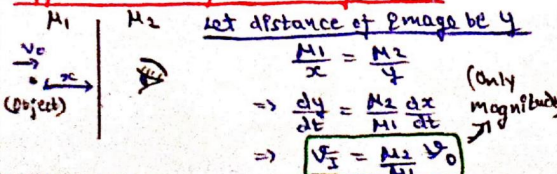
Object in Rarer medium



$$h_e = hM$$

$$\text{Shift} = h(M - 1)$$

Apparent Velocity due to Refraction:



$\rightarrow$ If observer is also moving, then apply concept of relative velocity.

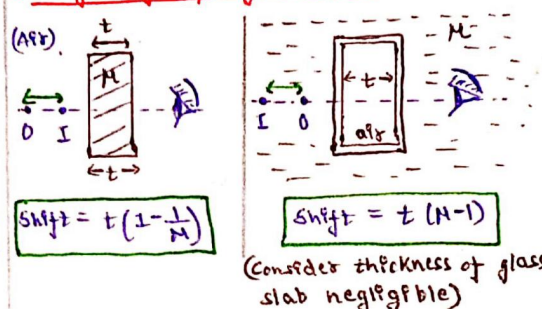
Eg: (A) (B) What is app. velocity of fish seen by bird if it is moving $\downarrow$ by v_B .

$$\text{Sol}^n \quad v_{FB} = \left(\frac{M_W}{1}\right)v_F + v_B$$

(P) What is app. velocity of bird observed by fish if bird is moving $\uparrow$ by v_B .

$$\text{Sol}^n \quad v_{BF} = \left(\frac{M_W}{1}\right)v_B - v_F$$

Image shift by a glass slab:



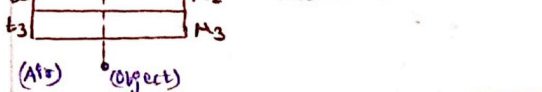
$$\text{Shift} = t \left(1 - \frac{1}{M}\right)$$

$$\text{Shift} = t(M - 1)$$

(Consider thickness of glass slab negligible)

Image shift due to multiple glass slab:

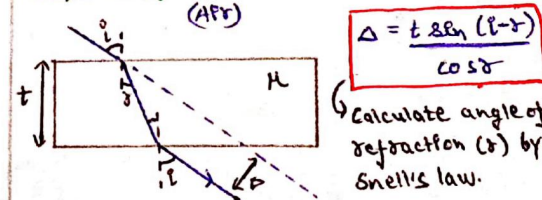
$\rightarrow$ Valid only for near normal / paraxial incidence.



$$\text{Total shift} = t_1 \left(1 - \frac{1}{n_1}\right) + t_2 \left(1 - \frac{1}{n_2}\right) + t_3 \left(1 - \frac{1}{n_3}\right)$$

Lateral Displacement by Glass Slab:

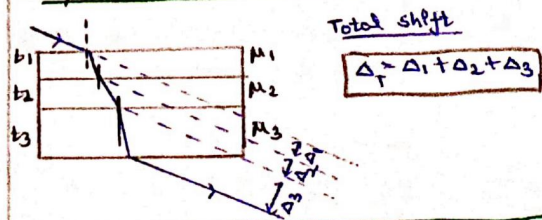
1) By a single Glass Slab $\Rightarrow$



$$\Delta = t \frac{\sin(\theta_1 - \theta_2)}{\cos \theta_2}$$

Calculate angle of refraction (θ_2) by Snell's law.

2) By multiple connected slabs $\Rightarrow$



Total shift

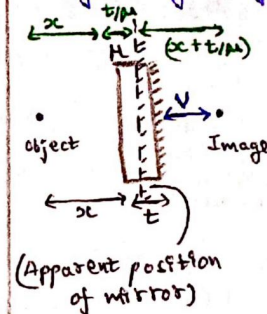
$$\Delta_T = \Delta_1 + \Delta_2 + \Delta_3$$

$$\Delta_T = t_1 \frac{\sin(\theta_1 - \theta_2)}{\cos \theta_2} + t_2 \frac{\sin(\theta_2 - \theta_3)}{\cos \theta_3} + t_3 \frac{\sin(\theta_3 - \theta_4)}{\cos \theta_4}$$

Subst individual displacements KP calc mean incidence angle first surface wala he use Karni hal.

Image Formation by a thick Mirror:

$\rightarrow$ In case of thick mirror, simply solve considering the apparent distance of reflecting surface.



$$\text{Distance b/w O \& I} = 2 \left(x + \frac{t}{M}\right)$$

$$\text{Actual distance of Image (v)} = x + \frac{t}{M} - t \left(1 - \frac{1}{M}\right)$$

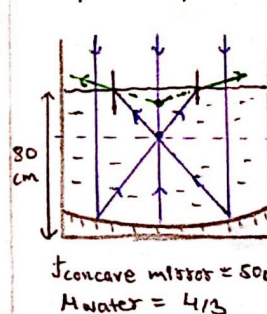
(Apparent position of mirror)

Effect of medium on Reflection:

$\rightarrow$ No effect on reflection $\Rightarrow$ no change in focal length of reflecting surface.

$\rightarrow$ Presence of medium only causes refraction.

Eg: Find image position, produced by this optical system for sunlight.



Sol: Due to sunlight, parallel rays from ∞ .

$\rightarrow$ Forms image at f (i.e. 50cm)

$\rightarrow$ Formed image serves as object for refraction

Concave mirror = 50cm
 $M_{\text{water}} = 4/3$

$$\text{So, } h_e = \frac{30}{M_0} = \frac{30 \times 3}{4}$$

$$\text{Ans } h_e = 22.5 \text{ cm}$$

Refraction at Spherical Surfaces:

For refraction at spherical surfaces,

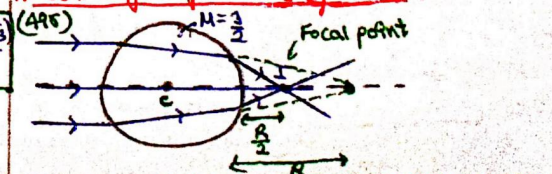
$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$n_1 =$ RI of incident
 $n_2 =$ RI of Refracted
 $R =$ Radius of surface

$$\text{Lateral magnification } m = \left| \frac{n_1 v}{n_2 u} \right|$$

For inverted or erect image, use
 O & I of same nature $\Rightarrow$ opp. side of P.A.
 O & I of opp. nature $\Rightarrow$ same side of P.A.

Focal length of a Glass Sphere:



For 1st Refraction $\Rightarrow$

$$u = \infty, \quad \frac{1}{v} - \frac{1}{\infty} = \frac{1}{R}$$

$$M_1 = 1, \quad v = \frac{1}{\frac{1}{R}} = R$$

$$M_2 = 3/2, \quad R = -R$$

is At R distance from second side of sphere.

For 2nd Refraction $\Rightarrow$

$$u = +R, \quad \frac{1}{v} - \frac{1}{R} = \frac{1}{-R}$$

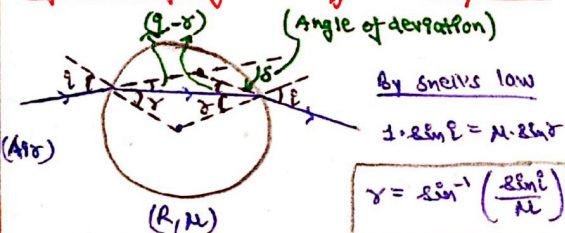
$$M_1 = M = 3/2, \quad \Rightarrow \boxed{v = +\frac{R}{2}}$$

$$M_2 = 1, \quad R = -R$$

is Final Image obtained at $\frac{R}{2}$ From First end
 $\frac{R}{2}$ From Second end.

Since Incident rays were parallel, so their Image will be formed at focus only.

Refraction of light through Glass sphere:

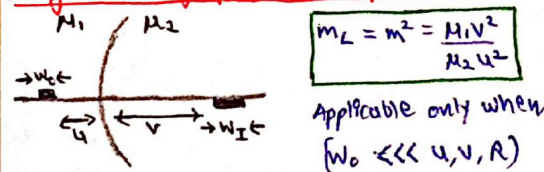


Angle of Deviation $\Rightarrow$

$$\delta = 2i - 2r = 2i - 2 \sin^{-1} \left(\frac{\sin i}{n} \right)$$

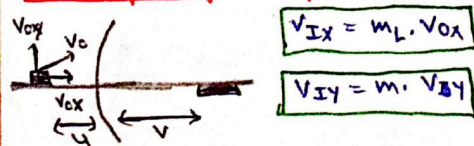
The analysis of Δ s done in above ray diagram is useful for refraction at other surfaces.

Longitudinal Magnification in Refraction:



Also, $m_L = \frac{w_i}{w_o}$

Velocity of Image in Refraction:

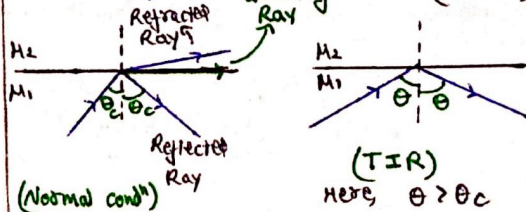


Total Internal Reflection (TIR)

Occurs only when a light ray travels from denser to rarer medium.

Tell critical angle (θ_c), along with reflection, refraction, some light still reflects into same

medium and some are grazing rays (along the interface).



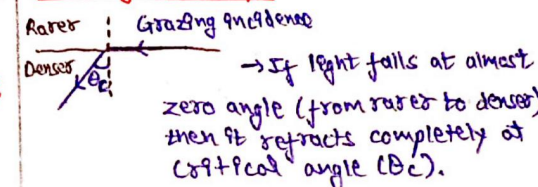
→ If the incidence angle is slightly more than θ_c , then all light is reflected.

Using Snell's Law

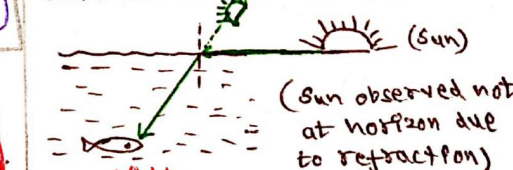
$$\theta_c = \sin^{-1} \left(\frac{M_2}{M_1} \right) = \sin^{-1} \left(\frac{1}{n} \right)$$

(For any medium) (For air)

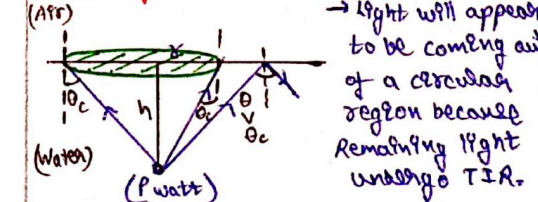
Grazing Incidence:



→ Therefore, even at sunset, sun light reaches to entire sea & sea-creatures can't see sun at horizon.



Point Light Source Inside Denser med.:



$$P_{esc} = \frac{P}{2} \left(\frac{1 - h}{\sqrt{1 + h^2}} \right)$$

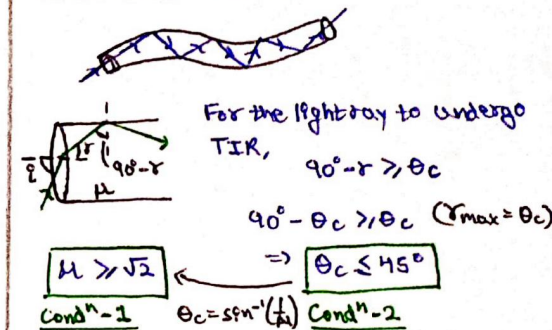
$$P_{esc} = \frac{P}{4\pi} \cdot 2\pi (1 - \cos \theta_c)$$

is Point light source embedded at centre of cube of side length 'a'. How much area should be covered to avoid it glow.

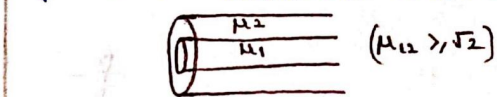


Optical Fibre:

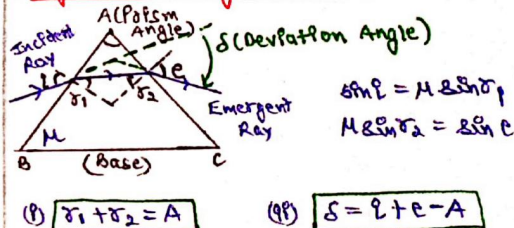
→ Fibre cables in which light travels by consecutive TIRs.



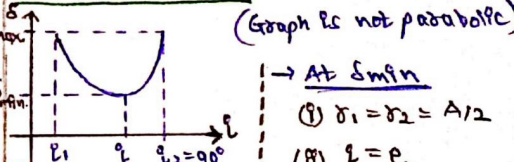
→ If medium of optical fibre is not capable to maintain $\theta_c \leq 45^\circ$ then 'Cladding' of some other material is done on it.



Refraction through a Triangular Prism:



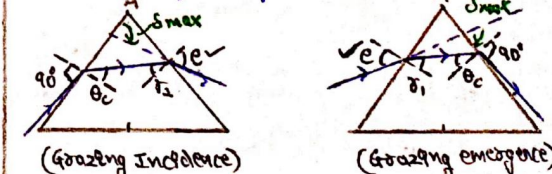
Minimum Deviation:



(3) Light passes through prism symmetrically

Maximum Deviation:

→ It happens in case of grazing incidence or grazing emergence.



Calculation of delta_max:

- 1) $r_2 = A - \theta_c$ → Get value of r_2
- 2) $\sin r_1 = n \sin (A - \theta_c)$ → Get value of e
- 3) $\delta_{max} = i + e - A$ ($i = 90^\circ$)

Condition for light to pass through Prism:

- 1) $A < 2\theta_c$
- 2) 'i' should be such that $r_2 < \theta_c$

Refraction by a Thin Prism:

→ For almost near normal incidence, passage of light is symmetric.

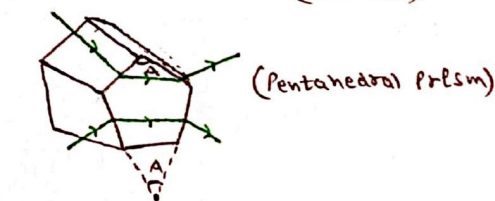
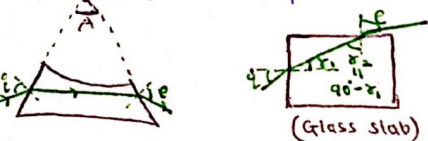
Since, $\theta \rightarrow 0$
 $\Delta \theta \approx \theta$

$$\Rightarrow \delta = A(\mu - 1)$$

Refraction through a Multi-faceted Prism:

→ Only the two surfaces on which successive refraction occurs are useful, rest are insignificant.

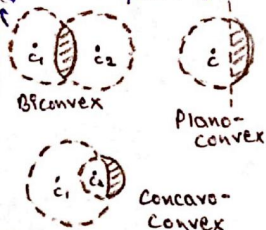
→ Find out angle b/w those two faces, treat it as prism angle (A) & solve assuming it as a bi-faceted prism.



Thin Lenses:

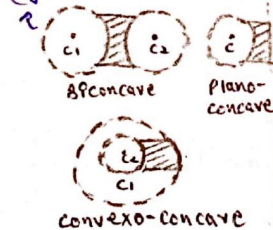
Convex Lenses (Converging)

$R_1 = R_2 \Rightarrow$ Equi-convex



Concave Lenses (Diverging)

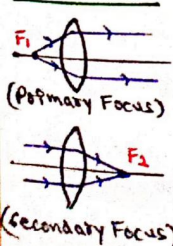
$R_1 = R_2 \Rightarrow$ Equi-concave



→ Focal length of a thin lens is independent from direction of incidence.

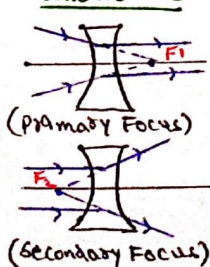
Primary & Secondary Focus of a Lens:

Convex Lens



For thin lenses, $f_1 = f_2$

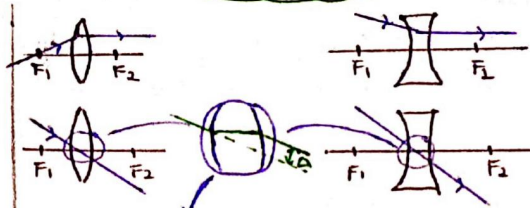
Concave Lens



Refraction through Thin Lenses:



Ray Optics - 3

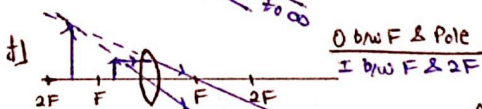
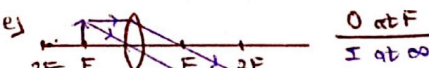
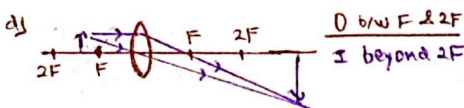
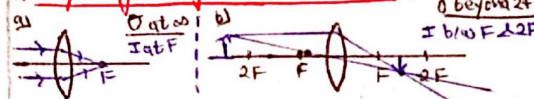


→ At centre, the surfaces of lens are almost parallel & lens behaves like a slab.

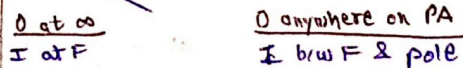
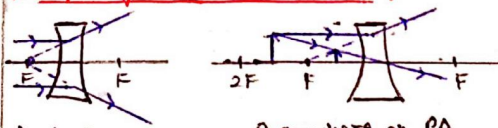
→ Thus, there is no deviation, but there is deflection of light at centre.

→ But, for thin lenses, it is very small, hence it can be neglected.

Ray Diagrams for Convex Lens:



Ray Diagram For Concave Lens:



Lens Maker's Formula:

→ This formula gives a value of focal length of a lens.

$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \quad (\text{App in surrounding})$$

$$\frac{1}{f} = \left(\frac{\mu_L}{\mu_S} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \quad (\text{Some other medium in surrounding})$$

$\mu_L \rightarrow$ RI of lens
 $\mu_S \rightarrow$ RI of surrounding medium

1) For Bi-concave / Bi-convex Lenses:

$$|f| = \frac{R_1 R_2}{(\mu - 1)(R_1 + R_2)}$$

$$|f| = \frac{R}{2(\mu - 1)} \quad (\text{When } R_1 = R_2)$$

2) For Plano-Convex / Plano-Concave Lens:

$$|f| = \frac{R}{\mu - 1}$$

3) For Convexo-Concave / Concavo-Convex Lens:

$$|f| = \frac{R_1 R_2}{(\mu - 1)(R_1 - R_2)}$$

Lens Formula & Magnification Formula:

Lens Formula $\Rightarrow$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Lateral Magnification $\Rightarrow$

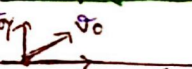
$$|m| = \left| \frac{v}{u} \right| = \frac{h_2}{h_1}$$

Longitudinal Magnification $\Rightarrow$

$$m_L = \frac{v^2}{u^2} = m^2 = \frac{W_2}{W_1}$$

(Only valid when $W_0 \ll u, v, f$)

Velocity of Image:



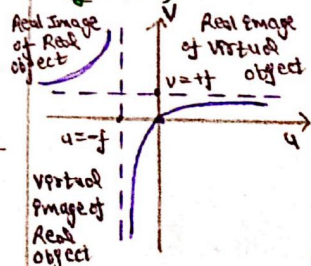
$$V_{ix} = m \cdot V_{ox}$$

$$V_{iy} = m_L \cdot V_{oy}$$

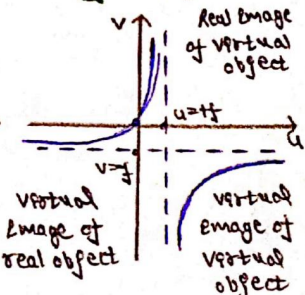
PA of any lens

Variation Curves of u & v for Lens:

Converging Lenses ($f = +ve$)



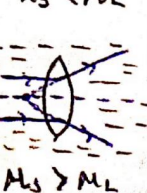
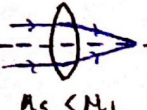
Diverging Lenses ($f = -ve$)



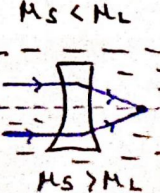
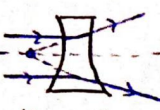
Converging & Diverging Nature of Thin Lenses:

→ Behaviour of lenses depends on the medium of their surrounding.

Convex Lens



Concave Lens

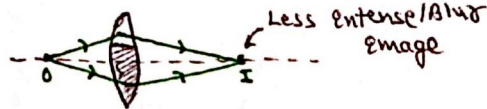


Cutting of Lenses:

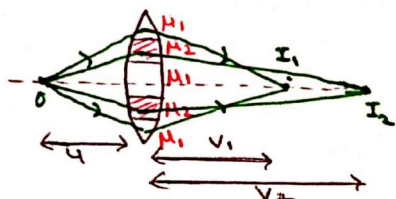
→ Every small part of lens behaves like a complete lens itself.

→ Even if some part of lens is cut or covered, then also remaining part forms image at the same location.

→ The only diff. is that its intensity ↓



→ If lens is made up of diff. materials, then it forms multiple image.



Calculate u, v_1 & v_2 by Lens maker's formula by same values of R_1 & R_2 .

Optical Power & Lens Mirror Combination:

Put f in metre
Unit of Power = Diopter (D)

$$P = \frac{1}{f}$$

For converging device → $P = +ve$

For diverging device → $P = -ve$

→ For combination of lenses & mirror, powers are added up.

Lens Combination ⇒

$$P_{eq} = P_1 + P_2$$

$$\Rightarrow \frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$$

Lens mirror combination ⇒

$$P_{eq} = P_L + P_M + P_L$$

Jointly both refraction hogi
Kesi lens se, unki baar uski
power add hogi

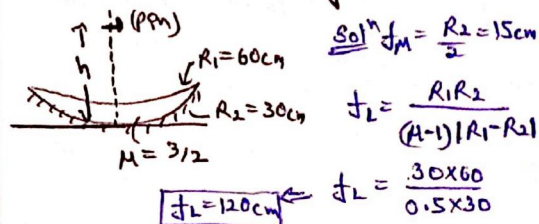
$$\Rightarrow \frac{1}{f_{eq}} = \frac{1}{f_M} + \frac{2}{f_L}$$

Image Formation at Location of object:

→ Whenever image of an object is formed at the position of object itself; then light must have had normal (low) incidence on the surface of optical device.

→ Also, the rays must pass through centre of curvature / optical centre of the device.

Eg: Find 'h' so that image of pin is produced on itself.



So,

$$\frac{1}{f_{eq}} = \frac{1}{f_M} + \frac{2}{f_L} \Rightarrow \frac{1}{f_{eq}} = \frac{1}{15} + \frac{2}{120}$$

$$\frac{1}{f_{eq}} = \frac{5}{60} \Rightarrow f_{eq} = 12 \text{ cm}$$

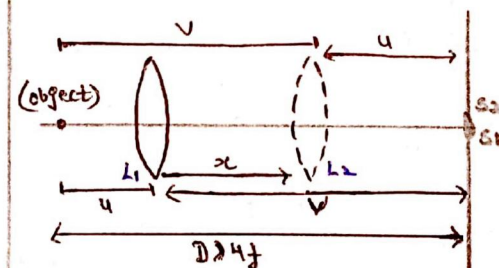
Image must be centre of curvature

$$\Rightarrow h = 2 \times f_{eq} = 24 \text{ cm Ans}$$

Displacement Method Experiment:

→ This experiment is carried out to find the focal length of a convex lens.

NOTE: For obtaining real image of a real object by a convex lens, the separation b/w object & image / screen must be $2 > 4f$



→ First lens kept at some 'u', from where sharp image of size s_1 obtained

→ Then we keep lens at 'v' of s_1 & obtain another sharp image of size s_2 .

$$D = u + v$$

$$x = v - u$$

$$f = \frac{D^2 - x^2}{4 \cdot D}$$

Size of object $s_0 = \sqrt{s_1 s_2}$

Geometric mean of sizes of both images

Magnification

For L_1 , $m_1 = |v_1/u|$
For L_2 , $m_2 = |v_2/v|$

$$m_1 \cdot m_2 = 1$$

Dispersion of Light:

→ It is splitting of white light into 7 constituent colours (VIBGYOR)

Cauchy's Relation

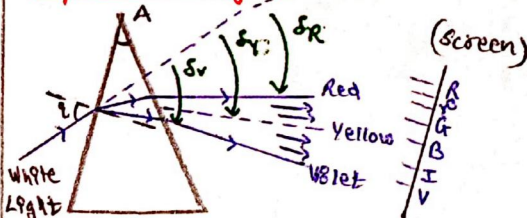
For different wavelengths of light, refractive index of the same substance is diff.

$$\mu = A + \frac{B}{\lambda^2} \quad (A, B \rightarrow \text{const})$$

Means, $\lambda_{\text{Red}} > \lambda_{\text{Violet}}$

$$\Rightarrow \mu_{\text{Red}} < \mu_{\text{Violet}}$$

Dispersion through prism:



→ Yellow light is taken as the mean ray & S_y is the mean.

Dispersion Angle, $D = S_v - S_R$

For small angled prisms ⇒

$$S_R = A(\mu_R - 1) \quad S_y = S_{\text{mean}} = A(\mu_y - 1)$$

$$S_v = A(\mu_v - 1)$$

Dispersive Power of a material (Prism) ⇒

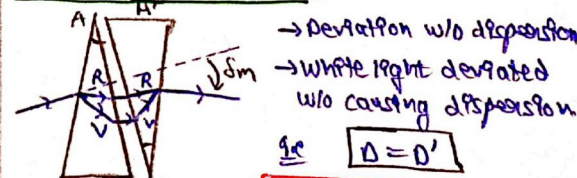
$$\omega = \frac{d\theta}{\theta} = \frac{\Delta\theta}{\theta} = \frac{\mu_v - \mu_R}{\mu_y - 1}$$

$\Delta\theta$ → Dispersion Angle

θ → Mean Deviation

Small Angled Prism Combinations:

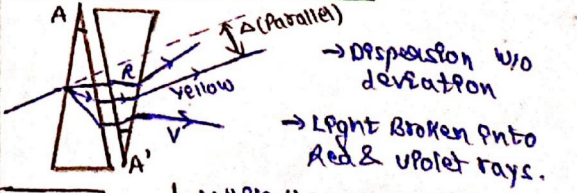
Achromatic Combination ⇒



$$A(\mu_v - \mu_R) = A'(\mu'_v - \mu'_R)$$

$$S_m = S_1 - S_2 = A(\mu_y - 1) - A'(\mu'_y - 1)$$

Direct Vision Combination ⇒



$$S_m = S'_m$$

$$A(\mu_y - 1) = A'(\mu'_y - 1)$$

$$D_{\text{Total}} = D + D' = A(\mu_v - \mu_R) + A'(\mu'_v - \mu'_R)$$